Chapter 8

Quantum randomness certification

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Exercises

Problem 8.1: Properties of entropies

We recall the definitions of the Shannon and min-entropy. The *Shannon entropy* measures the average amount of information (or equivalently the average unpredictability) in a random variable X with probability distribution $p(x)$. It is defined as

$$H(X) = -\sum_x p(x) \log p(x). \tag{8.1}$$

The *min-entropy* is the most conservative measure of information or unpredictability in a random variable. It is defined as the information content of the most likely event

$$H_{min}(X) = -\log\left(\max_x p(x)\right). \tag{8.2}$$

(a) Given a random variable X with d possible outcomes and distribution $p(x)$, prove that the Shannon entropy is maximised for the uniform distribution where all outcomes are equally likely $p(x) = \frac{1}{d}$.

(b) Prove that $H_{min}(X) \leq H(X)$ for any random variable X. (You may assume that X has a finite number of outcomes). When is $H_{min}(X) = H(X)$?

Consider now a random variable X with outcomes in $\{1, \ldots, d\}$ and distribution $p(x)$. Consider further a random mixing of the outcomes in which outcome l is replaced by k with probability μ_{kl}. This results in a new random variable X' again with outcomes in $\{1, \ldots, d\}$ but with a modified distribution

$$p'(k) = \sum_l \mu_{kl} p(l),$$

where $\mu_{kl} \geq 0$ and $\sum_k \mu_{kl} = 1$ (i.e. μ_{kl} defines a stochastic matrix).

(c) Show that $H(X') \geq H(X)$. When is $H(X') = H(X)$?

Problem 8.2: Estimating extractable randomness

Consider a classical random variable taking values in $\{0, 1\}$, representing a biased random bit with distribution

$$p(0) = q, \qquad p(1) = 1 - q. \tag{8.3}$$

Assume that the bias is known but no other side information is available.

(a) Compare the asymptotic number of extractable bits per round as given by the min-entropy with the number of bits extracted per round by the Von Neumann extractor, as functions of q. Make a plot.

Consider now a simple QRNG device consisting of a single-photon detector measuring a coherent state, as shown in Fig. 8.1. The coherent state has amplitude α. The detector discriminates only between no and one or more photons, and we label the no-click and click events by 0 and 1, respectively. The detector has non-unit efficiency $\eta < 1$. We can make a simple model of the device, enabling us to compute extractable randomnes, as follows.

We assume the behaviour of the device to be independent and identically distributed (i.i.d.) in separate rounds of the experiment. We associate a random variable X with the detector output and a random variable W with the side information available to an adversary. For an ideal detector with unit efficiency, the no-click event would occur with probability $r = |\langle 0|\alpha\rangle|^2 = e^{-|\alpha|^2}$. We make the (worst-case) assumption that all apparent inefficiency is due to the adversary. We take W to be a binary variable such that for $W = 0$, the detector never clicks and the adversary has perfect knowledge of the outcome, while for $W = 1$ it behaves as a perfect detector. The distribution of W is

$$P_W(0) = 1 - \eta, \qquad P_W(1) = \eta, \tag{8.4}$$

and the conditional distribution of X given knowledge of the side information W is

$$\begin{aligned} P_{X|W}(0|0) &= 1, & P_{X|W}(1|0) &= 0, \\ P_{X|W}(0|1) &= 1 - r, & P_{X|W}(1|1) &= r. \end{aligned} \tag{8.5}$$

(b) Compute the min-entropy for the model of the coherent-state QRNG given by Eqs. (8.4) and (8.5). How does it behave as a function of r and η? What is the best choice for r maximising the amount of randomness for a given efficiency? How many almost random bits with security parameter $\epsilon = 2^{-50}$ can be extracted from an initial raw string of 10^3 bits if the detector efficiency is 60%?

(c) Make a model of the QRNG from the previous exercise which includes both detector inefficiency and detector dark counts. Dark counts are events where the detector clicks regardless of whether or not any photons are present. You can assume that the adversary has full knowledge about when dark counts occur. Compute H_{min} for your new model.

Figure 8.1: A non-number resolving single-photon detector is used to measure a coherent state emitted by a laser source.

Problem 8.3: White-hat Eve

At random.jonatanbohrbrask.dk and rocket.jonatanbohrbrask.dk you find two games challenging the user to generate random bits while the computer attempts to predict the input.

Reverse engineer the games / look into the code to find out

(a) How does the algorithm generate its predictions? Why does it work?

(b) How is the quality of the input randomness evaluated? How is the score computed?

(c) Can you come up with an optimal strategy beating the computer?

Problem 8.4: Toeplitz hashing

One example of a quantum-proof strong randomness extraction method is hashing with random Toeplitz matrices. A Toeplitz matrix is a matrix with constant diagonals, i.e. any $l \times m$ matrix of the form

$$T = \begin{bmatrix} r_0 & r_1 & r_2 & \cdots & \cdots & r_{m-1} \\ r_m & r_0 & r_1 & & & \vdots \\ r_{m+1} & r_m & r_0 & & & \\ \vdots & & & \ddots & & \\ r_{m+l-2} & \cdots & & & & \end{bmatrix}. \tag{8.6}$$

When the first row and column are given, T is completely fixed. This requires specifying the $m+l-1$ numbers $r_0, \ldots r_{m+l-2}$. In Toeplitz hashing, a uniform random seed is used to pick bits $r_0, \ldots r_{m+l}$ and hence fix T. Organising the raw data in a (column) vector $\mathbf{x}$ of length m, extraction is performed by matrix multiplication, returning the extracted string $\mathbf{z} = T\mathbf{x}$ of length l (where addition of bits in the matrix multiplication is understood to be modulo 2 such that the output is again a bit string). This defines a two-universal hash family, and hence a quantum-proof strong extractor.

Note that the required seed is longer than the extracted string, $m+l-1 > l$, so in one round of extraction more randomness is consumed than extracted. That seems pretty useless! Indeed, if the extractor is applied to the entire string of raw data in one go, that would be the case - we would loose more than we gain. However, because the extractor is strong, the seed can be reused, which saves the day. Given raw data of length N, we can divide it into shorter blocks of length m and apply the extractor to each block using the same T, i.e. using the seed only once. The total extracted string can then exceed the seed length, even though each single block extracts less than the seed. In this exercise, we will explore the optimal block length.

Assume that we are given a total of N bits of raw data with a min-entropy per bit of h (i.e. the min-entropy of the raw data is Nh). We are also given a target security parameter of $\epsilon = 2^{-\mu}$. That is, the extracted distribution should be ϵ-close to uniform. We divide the data into N/m blocks of length m (you can assume that N/m is an integer), and pick an extraction length l. We then use a uniform seed of $m+l-1$ bits to generate an $l \times m$ Toeplitz matrix T and apply it sequentially to each block to perform extraction.

(a) Argue that, in the first round, the extracted string will be ϵ_1-close to uniform, with $\epsilon_1 = 2^{-\frac{1}{2}(mh-l)}$.

Each round of extraction leaks a small amount of information about the seed. Assume that, in the k'th round, the extracted string will be ϵ_k-close to uniform with $\epsilon_k = k\epsilon_1$.

(b) Argue that, for given block length m, the extraction length l should be at most $mh - 2\mu - 2\log_2(\frac{N}{m})$.

(c) The gain is the ratio of the number of extracted to seed bits. The optimal block length is the one maximising the gain. Using numerical optimisation, explore how the optimal block length depends on h, N, and μ. *NB.*

Determining the optimal block length analytically is possible but tricky and requires some careful approximations.

Problem 8.5: CHSH inequality and CHSH game

Consider the simplest Bell scenario with two parties (Alice and Bob), with binary inputs $x, y \in \{0, 1\}$ and outputs $a, b \in \{0, 1\}$. The CHSH Bell inequality on standard form is

$$S = \langle A_0 B_0 \rangle + \langle A_0 B_1 \rangle + \langle A_1 B_0 \rangle - \langle A_1 B_1 \rangle \leq 2, \tag{8.7}$$

where the correlators are defined in terms of the observed, conditional probabilities $P(ab|xy)$ as

$$\langle A_x B_y \rangle = P(a = b|xy) - P(a \neq b|xy) \tag{8.8}$$

$$= P(00|xy) + P(11|xy) - P(01|xy) - P(10|xy). \tag{8.9}$$

The 2 on the right-hand side of (8.7) is the local bound, and the inequality must be fulfilled by all probability distributions which allow a local-hidden-variable model. In quantum mechanics, the inequality can be violated, up to $S \leq 2\sqrt{2}$.

In the context of device-independent QRNG, it can be useful instead to work with a formulation of the Bell scenario in terms of a game. In the CHSH game, the goal of Alice and Bob is to produce outputs such that

$$a \oplus b = xy. \tag{8.10}$$

That is, if the inputs are $(x, y) = (0, 0)$, $(0, 1)$, or $(1, 0)$, then they should get $a = b$ and if the inputs are $(1, 1)$, then they should get $a \neq b$. The winning probability p_{CHSH} is the probability for the relation (8.10) to hold, when the inputs are provided uniformly at random.

(a) Prove that the CHSH inequality (8.7) implies that

$$p_{CHSH} \leq \frac{3}{4}, \tag{8.11}$$

and that the maximum quantum value $S = 2\sqrt{2}$ corresponds to $p_{CHSH} = \frac{1}{2}(1 + \frac{1}{\sqrt{2}})$. *Hint:* Use normalisation $\sum_{a,b} P(ab|xy) = 1$.

Problem 8.6: Quantum CHSH correlations

We can use tables of the following form to list all the probabilities $P(ab|xy)$ in the CHSH scenario in a compact manner.

$x\backslash y$		0		1	
	$a\backslash b$	0	1	0	1
0	0	$P(00\|00)$	$P(01\|00)$	$P(00\|01)$	$P(01\|01)$
	1	$P(10\|00)$	$P(11\|00)$	$P(10\|01)$	$P(11\|01)$
1	0	$P(00\|10)$	$P(01\|10)$	$P(00\|11)$	$P(01\|11)$
	1	$P(10\|10)$	$P(11\|10)$	$P(10\|11)$	$P(11\|11)$

One strategy to achieve the maximal quantum CHSH value is for Alice and

Bob to share the maximally entangled singlet state of two qubits

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle), \tag{8.12}$$

and perform projective Pauli measurements $\hat{A}_x = \boldsymbol{u}_x \cdot \hat{\boldsymbol{\sigma}}$ and $\hat{B}_y = \boldsymbol{v}_y \cdot \hat{\boldsymbol{\sigma}}$ with

$$\boldsymbol{u}_0 = \boldsymbol{e}_z \qquad \boldsymbol{v}_0 = -\frac{\boldsymbol{e}_x + \boldsymbol{e}_z}{\sqrt{2}}, \tag{8.13}$$

$$\boldsymbol{u}_1 = \boldsymbol{e}_x \qquad \boldsymbol{v}_1 = \frac{\boldsymbol{e}_x - \boldsymbol{e}_z}{\sqrt{2}}, \tag{8.14}$$

where $\boldsymbol{e}_x$, $\boldsymbol{e}_z$ are unit vectors along the x- and z-axis, respectively.

(a) Illustrate the measurements of Alice and Bob on the Bloch sphere.

(b) Prove that, for general Pauli measurements $\hat{A} = \boldsymbol{u} \cdot \hat{\boldsymbol{\sigma}}$ and $\hat{B} = \boldsymbol{v} \cdot \hat{\boldsymbol{\sigma}}$, the correlator in the singlet state is $\langle \Psi^- | \hat{A} \otimes \hat{B} | \Psi^- \rangle = -\boldsymbol{u} \cdot \boldsymbol{v}$.

(c) Prove that the probability distribution corresponding to the strategy above (when associating outcomes 0,1 to the +1,-1 eigenvalues of the observables, respectively) has the form

$x\backslash y$		0		1	
	$a\backslash b$	0	1	0	1
0	0	$\frac{1}{2}-\epsilon$	ϵ	$\frac{1}{2}-\epsilon$	ϵ
	1	ϵ	$\frac{1}{2}-\epsilon$	ϵ	$\frac{1}{2}-\epsilon$
1	0	$\frac{1}{2}-\epsilon$	ϵ	ϵ	$\frac{1}{2}-\epsilon$
	1	ϵ	$\frac{1}{2}-\epsilon$	$\frac{1}{2}-\epsilon$	ϵ

and find ϵ.

Problem 8.7: Trace distance generalises statistical distance

For two probability distributions P_X, P_Y defined over the same domain V, the *statistical distance* (or total variation distance) between them is

$$D(P_X, P_Y) = \frac{1}{2} \sum_{v \in V} |P_X(v) - P_Y(v)|. \tag{8.15}$$

Consider two quantum states diagonal in the same basis

$$\hat{\rho} = \sum_{v=1}^{d} p_{\hat{\rho}}(v) |v\rangle\langle v|, \qquad \hat{\sigma} = \sum_{v=1}^{d} p_{\hat{\sigma}}(v) |v\rangle\langle v|, \tag{8.16}$$

where $\{|v\rangle\}$ is som d-dimensional, orthonormal set.

(a) Show that $T(\hat{\rho}, \hat{\sigma}) = D(p_{\hat{\rho}}, p_{\hat{\sigma}})$, where T is the trace distance as defined in Eq. (2.91).